Schooling and Labor Market Consequences of School Construction in Indonesia: Reply to Roodman (2023)

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Abstract

Roodman (2023) revisits Duflo (2001), which evaluates the impact of the Indonesia INPRES SD school construction program on education and wages. Duflo 2001 uses a difference-in-differences specification that compared younger and older cohorts across low- and high-intensity program districts. Roodman (2023) proposes a number of minor adjustments, which do not substantially affect the results. He also makes one substantive comment: if education is correlated with a steeper age profile of wages (especially early in one's career), because INPRES was targeted at low-education districts, this could lead to a spurious difference-in-differences when comparing younger and older cohorts. He suggests that the solution to this problem is to use data where all workers are older, at which point the age profile of wages should be parallel. However, when he implements this strategy with 2013–14 data, he finds no impact of the SD program on years of education, which is surprising but prevents him from using this analysis as an experiment to look at the impact of education of wages (in the absence of a first stage). In this note, we implement minor corrections to Roodman's analysis, and use his approach on earlier years of the SUSENAS (2011–2012). In 2011–2012, we find a first stage for education (though weaker than what was reported in 1995) and we find that the IV results are similar to those originally reported Duflo (2001), consistent with the findings in Hsiao (2023), which independently replicates Duflo (2001) using a slightly different approach.

Duflo (2001) analyzes the impact of the *Instruksi Presiden Sekolah Dasar* (INPRES SD) school construction reform in Indonesia on schooling and wages and uses the program as the basis for an instrumental variables estimate of returns to education. Simply comparing outcomes for districts that received many new schools (i.e., "high intensity") to those that received few ("low intensity") is problematic because assignment of program intensity was not random: INPRES was targeted at low-education districts. Comparing people who were young enough to have attended the new schools to those who were too old is also problematic because education levels are higher in younger cohorts regardless. This setting lends itself to a difference-in-differences strategy: program intensity in the district of birth (high vs. low) being the first difference, and cohort of birth (young vs. old) being the other. Duflo shows that the education gap between high- and low-intensity districts shrinks in the younger cohorts, which, under a parallel trends assumption — that, absent the program, the high- and low-intensity districts would have followed the same trends — implies a significant positive effect of school construction on educational attainment. She employs the same strategy for wages, finding a similar effect. Finally, she uses these results in an instrumental variables estimation of the returns to education, wherein INPRES SD serves as an exogenous instrument for education.

Since the original publication, a number of papers have reanalyzed the INPRES SD natural experiment and have incorporated various advances in econometrics, such as spatial clustering, auto-correlation over time, and heterogeneous effects in difference-in-differences (De Chaisemartin and D'Haultfœuille 2018). This data has also been used to study the impact of education on outcomes other than wages (Ashraf et al. 2020; Bazzi, Hilmy, and Marx 2020; Martinez-Bravo 2017), and on outcomes for people other than those directly affected (Duflo 2004; Mazumder, Rosales-Rueda, and Triyana 2019; Akresh, Halim, and Kleemans 2021; Hsiao 2023). Generally, the Duflo (2001) results have been shown to be qualitatively robust, though less precise than initially portrayed. However, in recent work, Roodman (2023) is much more critical of the substance of the findings, concluding that despite some evidence that INPRES SD increased schooling, there is no compelling evidence of an effect on wages.

Roodman (2023) proceeds in two steps. First, he highlights a number of relatively

minor or technical issues, which he refers to as "micro" comments. These include clustered standard errors, the inclusion of survey weights, data transcription errors, and weak instrument adjustments. Together, these do not change the results substantially, although they weaken their (apparent) precision — as seen in other refinements of the Duflo (2001) results (De Chaisemartin and D'Haultfœuille 2018).

Second, Roodman (2023) raises a deeper conceptual problem with the identification strategy: a "macro" comment. Roodman finds evidence that the education wage gap increases with age: wages increase sharply with age (in the cross-section) for highly educated young and middle-aged men, whereas for men with little education, wages are fairly flat with age. Since INPRES SD targeted low-education districts, when Duflo (2001) compares high-intensity to low-intensity districts, she inevitably finds a larger wage gap across districts within the older cohort than within the younger cohort — not due to any impact of the program, but due to natural increase in wages with age for high-education workers. Roodman refers to this as the "dilation problem."

Roodman suggests that a natural solution to this problem is to use data that follows up with these cohorts later in life; in the cross-section, the dilation problem attenuates and even reverses among older men. The challenge with this strategy is data availability. Duflo (2001) uses the 1995 Intercensal Population Survey (SUPAS): in 1995, the cohorts of interest are 21–38 years old, which is well within the range that Roodman suggests would be subject to the dilation problem. Encouragingly, Roodman identifies three data sets that follow up with this cohort recently enough that they should largely avoid the dilation problem: the 2005 SUPAS, the 2010 Labor Survey (SAKERNAS), and the 2013 and 2014 Household Socioeconomic Surveys (SUSENAS). However, the 2005 SUPAS does not have wage data, and despite Roodman's heroic attempts to impute the missing data, we contend that only a data set that includes wages can reliably demonstrate the effect of schooling on wages. Meanwhile, the 2010 SAKERNAS does not have region of birth; as Hsiao (2023) shows, migration is a driving factor behind the results, so imputing school-age residence from current residence is a risky endeavor. This leaves the SUSENAS as the only data set suitable for this analysis.

Unfortunately, Roodman encounters a problem in the 2013 and 2014 SUSENAS: the first stage relationship between program intensity and wages and years of education seems to have

mostly disappeared. As he points out, this may be due to recall bias or measurement error — which could affect educational attainment, district of birth, or year of birth: a suspiciously large fraction of the sample reports being born in a year ending in 0 or 5. This measurement error would introduce a downward bias in both the education and the wage reduced form and makes the IV non-interpretable. After correcting a typo in Roodman's construction of the years of schooling variable, we find an even weaker first stage. Thus, none of Roodman's followup data speaks to whether the effect of school construction on wages holds beyond 1995, leaving the dilation problem unresolved.

Coincidentally, in unrelated work (the early version of which predated Roodman (2023)), which he undertook as a motivation of his own study of the optimal education policy in a world with migration, Hsiao (2023) also reproduced part of the original Duflo (2001) analysis using the SUSENAS. In contrast to Roodman (2023), Hsiao uses four years of the SUSENAS (2011–14), and finds quite different results: applying the Duflo (2001) difference-in-differences specification to the new data, Hsiao find a positive impact of school construction intensity on education and wages. The results are similar in magnitude to those found in Duflo (2001), implying, if anything, slightly larger instrumental variable estimates. The constrast between the two papers is a little surprising.

Roodman, having not acquired this data and therefore being unable to resolve the dilation problem, proposes other approaches. First, he formally tests for trend breaks; supposing that the difference between high- and low-intensity regions follows a trend as the progress from one cohort to the next, any impact of INPRES SD would result in a kink in that trend. If we consider the dilation problem to be a form of parallel trends violation, then testing for trend breaks would effectively control for the pre-trends. The result of the trend-break analysis in Roodman (2023) are somewhat inconclusive: clear breaks appear for wages in the 1995 SUPAS, but not in the SUSENAS 2013–2014. Further, these are not robust to controlling for a quadratic function of the age cohort.

Finally, because wage dilation should have no effect on wage rank, Roodman borrows the changes-in-changes estimator from Athey and Imbens (2006), finding little evidence of positive impact of INPRES SD — which contrasts with the strong changes-in-changes result found in Hsiao (2023). He also discusses why the generalization of Athey and Imbens (2006)

developed by De Chaisemartin and D'Haultfœuille (2018) with their fuzzy difference-in-differences approach, which finds positive impacts of INPRES on education and wages, is inappropriate for this setting.

In this note, we respond to both sets of Roodman's comments. To address his micro comments, we reproduce his analysis, updating the main estimates in Duflo (2001). We apply Roodman's corrections to the data set, which we have thoroughly vetted and fully endorse, and corroborate his findings that the 1995 results are of similar magnitudes but are less precise than reported in Duflo (2001). To address his macro comments (the "dilation problem"), we reproduce and then build upon his analysis of the 2013 and 2014 SUSENAS. When we combine the 2011 and 2012 SUSENAS with Roodman's 2013–14 data and correct Roodman's coding errors, we find results that are largely consistent with those of Duflo (2001). This is almost entirely driven by the 2011–12 data: in the unweighted regressions,[1] when we isolate 2011–12, we find that school construction intensity is correlated with large, positive, and highly significant increases in educational attainment and income; but in 2013–14, the waves used by Roodman, the results on education are small and insignificant, which precludes running instrumental variables regressions. In fact, the first stage is strongest in 2011.

Overall, the positive and moderately significant results (wherever we find an impact on education) in this recent follow-up survey of cohorts first studied in Duflo (2001) lend little credence to the "dilation" hypothesis, and suggest that Duflo's original results stand. The difference between 2011–12 and 2013–14 is surprising, but given that a first stage was found in different data set by different papers, this is perhaps likely to be a fluke in the 2013–2014 data.

1. Surprisingly, there are large effects of weighting in this data set — including survey weights renders the first stage too weak to be of any use. As we explain below, we do not think weights should be used when trying to find the causal effect of education on wages. Replication code for all of the analysis here is available at https://github.com/NathanLazarus/ReplyToRoodman.

1 Setting and basic empirical strategy

In 1973, the central government of Indonesia began a set of development projects under the name of *Instruksi Presiden* ("presidential instructions," INPRES). The largest of these programs was INPRES *Sekolah Dasar* ("elementary school," SD), which resulted in the construction of over 60,000 primary schools within six years, doubling the stock of primary schools in the county. This was not only an infrastructure investment: in order to staff the schools, the government ramped up training and recruiting of teachers, increasing their stock of teachers by 43 percent (Jellinek 1992).

The schools were meant to be targeted at districts where a lower share of primary school–age children are enrolled.

Due to the nonrandom assignment of the program, Duflo (2001) uses a difference-in-differences strategy to examine how outcomes differ across age cohorts as a function of program intensity. In order to provide intuition, she illustrates the most basic version of this approach using two binary variables: INPRES SD intensity (high vs. low, i.e., "treatment" vs. "control") and age cohort ("young" vs. "old"). The "old" cohort was 12–17 years old in 1974: too old to benefit from primary school construction. Comparing their outcomes to those of the "young" cohort (age 2–6 in 1974 — exactly the right age to possibly have benefited from the new schools) is the first difference. Doing so separately within both high- and low-intensity regions and then comparing those differences yields the difference-in-differences estimate.

Table 3 of Duflo (2001) shows these basic two-by-two difference-in-differences results. She finds that education levels increases by 0.36 years in the low-intensity regions between the old and young cohorts, and by 0.47 years in the high-intensity regions: subtracting the former from the later give an estimated effect of INPRES SD of 0.12 years of schooling. Following the same procedure for wages, she finds an effect of about 2.6%. Dividing the wage result by the education result gives us a Wald estimate of the returns to education for compliers (i.e., those who were induced to increase their education by the high-intensity program): each additional year of schooling increases wages by over 20%, in this case.

This empirical strategy relies on the parallel trends assumption: that the evolution of

outcomes across cohorts in low-intensity districts serves as a valid counterfactual for the evolution of outcomes across cohorts in high-intensity regions. This is not obviously true. Perhaps low-education areas are naturally converging towards high-education districts over time, in which case we would expect the gap between high- and low-intensity districts to be smaller in the young cohorts even absent INPRES. To bolster the claim of parallel trends, (Duflo 2001) constructs a placebo experiment where she compares the old cohort to an even older cohort (age 18–24 in 1974), neither of which should have benefited from INPRES. If high-intensity and low-intensity districts were converging even without INPRES, then if we apply the difference-in-differences strategy to these two cohorts, we should also expect to find an effect. Reassuringly, in her Table 3, Duflo (2001) finds a zero in this placebo experiment.

These results, both the experiment of interest and the placebo experiment, are somewhat noisy. By condensing the data into two binary variables, Duflo throws away much of the variation that may be useful for more precisely estimating the effect of INPRES SD. To recover some of that lost precision, Duflo (2001) deploys the following regression specification:

$$S_{ijk} = \alpha_j + \beta_k + (P_j T_i)\,\gamma + (\mathbf{C}_j T_i)\,\delta + \varepsilon_{ijk}, \tag{1}$$

where S_{ijk} is schooling for person i in born in district j in year k, α_j is district fixed effects, β_k is birth year fixed effects, P_j is INPRES SD intensity (number of new primary schools constructed per 1000 children in district j), T_i is a "young" cohort indicator (age 2–6 in 1974), and $\mathbf{C}_j$ is a vector of district-specific controls (which consist of three sets of interactions between birth-year dummies and the three following district j–specific variables: number of children, primary school enrollment rate, and INPRES water and sanitation spending per capita). Duflo (2001) replaces S_{ijk} with Y_{ijk} when estimating the effect of INPRES SD on (the logarithm of) wages.

While this specification uses a more continuous measure of program intensity and exposure, the idea behind the illustrative difference-in-differences still applies. Birth year fixed effects control for any cohort-specific shocks, but the comparison is still essentially between those in the "young" cohort ($T_i = 1$), which could have benefited from INPRES SD schools, and those too old to have done so. Treatment intensity (P_j) is now allowed to vary

continuously such that when interacted with T_i, it yields a coefficient of interest (γ) that represents the effect of one additional school per 1000 children — for those young enough to have possibly attended that school.

Duflo (2001) presents the results of this specification in Table 4. In Panel A, where the sample is restricted to those in the young or old cohorts, she finds significant positive effects of INPRES SD on both educational attainment and wages. Depending on the controls included, when she restricts the sample to wage-earners, she finds that one additional school per 1000 children is associated with 0.20–0.26 additional years of schooling and 1.47–2.70% higher hourly wages. In Panel B, she presents the placebo experiment for this same specification, restricting the sample to those in the old cohort and the older cohort, and finds no significant effect.

To examine how the effects of INPRES SD evolve across the cohorts, Duflo (2001) also runs a more flexible regression:

$$S_{ijk} = \alpha_j + \beta_k + \sum_{\tau=2}^{23} (P_j d_{i\tau})\, \gamma_\tau + \sum_{\tau=2}^{23} (\mathbf{C}_j d_{i\tau})\, \delta_\tau + \varepsilon_{ijk}, \tag{2}$$

where $d_{i\tau}$ is an indicator for whether person i was of age τ in 1974, and the other variables are defined as they were in equation (1): S_{ijk} is schooling for person i in born in district j and year k, α_j is district fixed effects, β_k is birth year fixed effects, P_j is INPRES SD intensity, and $\mathbf{C}_j$ is a vector of district-specific controls. Again, the outcome S_{ijk} can be replaced with Y_{ijk} to see the effect of INPRES SD on wages.

The results of this regression are presented in Duflo (2001) Figure 1. For ages 12–24, $d_{i\tau}$ (the coefficient of interest) fluctuates around 0, which is to be expected if INPRES SD is uncorrelated with any uncontrolled confounding variable that was present before INPRES SD took effect; i.e., there appear to be no pre-trends. For those younger than 12 years old in 1974, $d_{i\tau}$ increases steadily with youth. This is also to be expected: many children have dropped out of school by the time they reach 11 years old, so a new primary school does them no good. On the other hand, for the parents of a two-year-old, a new school makes the entire primary school experience suddenly look more appealing. In total, if the "cost" of schooling is lowered (i.e., improved access), those savings are multiplied by the number

of years a child is eligible to attend, so the effect of INPRES SD should be the greatest for those who were the youngest in 1974. Duflo (2001) finds a similar pattern in Figure 3, where she uses wages (Y_{ijk}) as the outcome of interest.

Finally, these results can be used to estimate the returns to education in an instrumental variables specification, where INPRES SD serves an exogenous instrument for schooling. To be specific, the first stage is equation (1) when the outcome of interest is schooling, and the reduced form is equation (1) when the outcome of interest is wages. The two-stage least squares specification is therefore the following:

$$Y_{ijk} = \alpha_j + \beta_k + \gamma_{2SLS}\widehat{S}_{ijk} + (\mathbf{C}_j T_i)\,\delta + \varepsilon_{ijk}, \tag{3}$$

where $\widehat{S}_{ijk}$ is the fitted values from equation (1), and the rest of the variables are defined as before: Y_{ijk} is income for person i in born in district j and year k, α_j is district fixed effects, β_k is birth year fixed effects, P_j is INPRES SD intensity, and $\mathbf{C}_j$ is a vector of district-specific controls.

Duflo (2001) also ran a two-stage least squares specification using a slightly modified version of equation (2) as the first stage — instead of using a dummy for each year of birth, she pools those aged 12–24 into one large control group, since, as she shows in Figures 2 and 3, the effect of the program is effectively zero in the pre-period:

$$S_{ijk} = \alpha_j + \beta_k + \sum_{\tau=2}^{12} (P_j d_{i\tau})\,\gamma_\tau + \sum_{\tau=2}^{12} (\mathbf{C}_j d_{i\tau})\,\delta_\tau + \varepsilon_{ijk}. \tag{4}$$

Setting aside issues having to do with weak instruments for now, this could have more statistical power compared to when equation (1) is used as the first stage. She presents the results of these two-stage least squares regressions in Table 7, finding an increase in wages of about 5–12% per additional year of schooling, depending on the exact specification.

2 "Micro" Comments

Roodman (2023) points out a number of small issues with the implementation of the empirical strategy in Duflo (2001). As Roodman says, many of these are simply a product of the fact

that much progress has been made in econometric techniques since 2001. We address them one by one here, updating the specifications discussed above where appropriate, and show those results in Table 1.

In the following subsections, we respond to the micro comments in detail.

2.1 Clustering

Roodman suggests that the standard errors are too small due to lack of appropriate clustering. The need to cluster at the unit of treatment to account for spatial correlation has been known since Bertrand, Duflo, and Mullainathan (2004). In this case, because INPRES SD assigns new primary schools at the district level, we must cluster by district of residence at the age of schooling. While the 1995 SUPAS does not include district of residence in childhood, it includes district of birth, which serves as a very good proxy, so that is the appropriate level of clustering.

To our knowledge, anyone who has used this setting for further analysis has indeed clustered the standard errors at the district level. This issue was, in fact, already pointed out in De Chaisemartin and D'Haultfœuille (2018), who found that after the appropriate clustering has been applied, the Duflo (2001) result "loses some statistical significance." We find the same. Comparing Column (2) to Column (1) in Table 1 illustrates exactly to what extent clustering changes the precision of the results. For example, it widens the 95% confidence interval of the first stage effect of school construction on years of schooling from $[0.113, 0.279]$ to $[0.100, 0.291]$ — hardly a dramatic change. However, the 95% confidence interval on the reduced form now (barely) includes zero.

2.2 Typos

Duflo hand collected the data in the Bappenas library, and a few errors were made in the process. For example, when recording the quantity of schools assigned to be constructed in each district each year, a few of those quantities were interchanged across consecutive districts. Roodman checks the entire INPRES SD data set against the government records, which have now been made public, and corrects them. We corroborate these corrections

— they are, indeed, faithful to the documents that Roodman obtained. In the future, all researchers studying INPRES should use the corrected data set.

Comparing Column (3) to Column (2) illustrates exactly to what extent the typos affect the results. For example, the first stage estimate decreases by 19% (from 0.196 to 0.159), while the reduced form increases by 27% (from 0.015 to 0.019). As a result, both second-stage estimates are actually substantially higher than what they were in Duflo (2001), increasing from 0.075 to 0.117 for the single-instrument specification (equation 3) and from 0.068 to 0.108 for the many-instrument specification (equation 4).

These changes also increase the standard errors slightly. Though the first stage estimate is still highly significant, the 95% confidence interval on the reduced form is now $[-0.003, 0.040]$. And although the second-stage estimates increase once the typos are corrected, the standard errors also increase approximately proportionally, so these estimates are no more statistically significant than they were in Duflo (2001).

2.3 Weighting

The Indonesian surveys oversample some people, and sampling weights are provided. Duflo (2001) uses the weights for descriptive statistics — in order to accurately describe the Indonesian population — but not in the regressions: weights are useful to compute averages that are representative of a specific population, but for regressions, when we are targeting a specific "structural" parameter, OLS is typically the "best linear unbiased estimator" (BLUE). Adding weights only increases variance. Furthermore, it is neither clear how people were sampled nor how these weights were constructed; therefore, and therefore what the sample is representative of when we use the weights. Some districts were more likely to be sampled by others, but the sampling weights also vary by district.

Roodman (2023) made a serious effort to understand the sample construction, but ultimately could not get precise answers. He writes, "Most likely sampling was stratified with respect to an index of household assets using data gathered in the 1990 census. The rarer and probably more diverse high-asset households would have been oversampled for precision." Roodman argues that the weights are therefore "endogenous," since people with more schooling and higher wages were more likely to be sampled, and provides evidence for

this claim by adding the survey weights as a control variable to equation (2) and finding "large, cluster-robust t statistics."

Note that the oversampling of richer people would only bias the estimates if it created a systematic correlation between the treatment (the interaction of new schools constructed with cohort) and the error term; this is not necessarily the case (Solon, Haider, and Wooldridge 2015). We thus continue to think that weights should not be used in this case (or in the later analysis), and that OLS is BLUE. However, we present results with and without weights.

Comparing Column (4) to Column (3) illustrates that weighting makes little difference to the estimates, although it (mechanically) increases standard errors. For example, when comparing the clustered, typo-corrected specification to the same specification with weights included, the coefficient on the first stage decreases by 3%, from 0.159 to 0.154, and the reduced form decreases 10%, from 0.019 to 0.017. However, including the weights increases the standard errors: the 95% confidence interval is now $[0.008, 0.301]$ for the first stage and $[-0.014, 0.048]$ for the reduced form.

2.4 Instruments

Since 2001, the literature on weak instruments has made great progress, and it is still an active area of research (Mikusheva and Sun 2022). It has become clear that the first stage in any instrumental variables specification must be strong to take the second-stage results seriously. In many of the instrumental variables studies published during this era, the instruments are likely too weak. This is a problem in particular for the two-stage least squares regressions in Duflo (2001) that use many instruments; the regressions using just one instrument should be somewhat less vulnerable to this pitfall. For this reason, as suggested by Roodman (2023), we include a test of instrument strength. Roodman suggests using the Kleibergen–Paap (KP) heteroskedasticity-robust first-stage F-statistic (Kleibergen and Paap 2006); however, the KP F-statistic is inappropriate except when the number of instruments is exactly one (Andrews, Stock, and Sun 2018). Instead, we use the Montiel Olea–Plueger (MOP) effective first-stage F-statistic (Montiel Olea and Pflueger 2013); this is true for any number of instruments, including the single-instrument case,

when the MOP and KP F-statistics are equivalent.

Roodman (2023) also cites the recommendation of Finlay and Magnusson (2019) to use the Anderson–Rubin (AR) test with a wild-bootstrapped distribution for the test statistic (Anderson and Rubin 1949). While the AR test is a good approach to the single-instrument specifications, as Roodman acknowledges, AR confidence intervals can be too wide with many instruments (Andrews 2016). The size of AR confidence intervals is correct only when the instruments would pass an over-identification test, while instruments in practice often identify different local average treatment effects. Despite cautions from econometricians in this regard (Andrews, Stock, and Sun 2018), the literature has yet to converge on better weak-instrument robust methods with many instruments; in particular, the approach of Mikusheva and Sun (2022) does not allow for clustering, otherwise we would apply it here. Therefore, we report the wild bootstrapped AR confidence sets, as suggested by Roodman, for both the single and many instrument specifications.

The F-statistics and 95% confidence sets in Panels C and D paint a clearer picture of instrument strength than did Duflo (2001). The single-instrument IV regressions of Panel C appear to be more robust than the many-instrument regressions of Panel D, although neither sets of first stages will be winning any Olympic gold medals in powerlifting. It is probably wise to take the highly-significant positive results of the many-instruments IV specifications with a grain of salt and focus more on the similarly large single-instrument IV specifications — or, better yet, on the reduced form results.

2.5 Overall Effect of Micro Comments

As illustrated in Table 1, these corrections made in Roodman (2023), though helpful, make little difference to the conclusion of Duflo (2001). The point estimates are remarkably stable even after all the corrections have been applied (see Column (4)), and though the standard errors have increased from what was originally reported more than 20 years ago, they do not meaningfully affect our understanding of the impacts of the school construction program on wages or on education.

Table 1: Micro corrections to Duflo (2001)

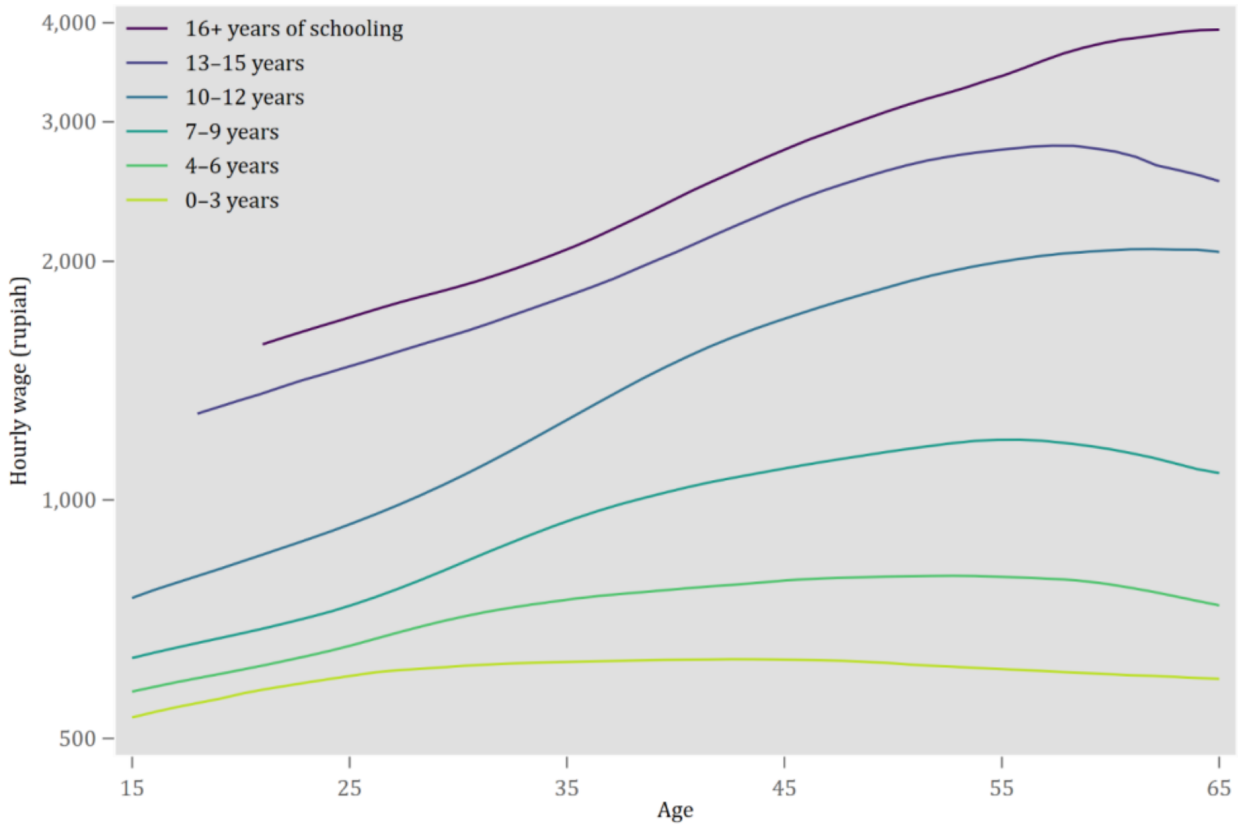
Outcomes	(1)	(2)	(3)	(4)
Panel A: First Stage (instrument: school construction interacted with "young" cohort dummy)				
Years of Schooling	0.196***	0.196***	0.159**	0.154*
	(0.042)	(0.048)	(0.058)	(0.075)
Panel B: Reduced Form (instrument: school construction interacted with "young" cohort dummy)				
log(Hourly Wages)	0.015*	0.015	0.019	0.017
	(0.007)	(0.009)	(0.011)	(0.016)
Panel C: 2SLS (instrument: school construction interacted with "young" cohort dummy)				
log(Hourly Wages)	0.075*	0.075	0.117*	0.109
	(0.034)	(0.040)	(0.058)	(0.090)
95% Confidence Set	$[0.004, 0.151]$	$[-0.019, 0.176]$	$[-0.030, 0.314]$	$[-0.504, 0.840]$
F-Stat	21.309	16.168	7.450	4.242
Obs.	31061	31061	31061	31061
Panel D: 2SLS (instruments: school construction interacted with birth-year dummies)				
log(Hourly Wages)	0.068*	0.068*	0.108*	0.136**
	(0.028)	(0.034)	(0.048)	(0.050)
95% Confidence Set	$[-0.060, 0.227]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$
F-Stat	2.701	0.331	0.171	0.337
Obs.	60663	60663	60663	60663
Clustering		✓	✓	✓
Typos Corrected			✓	✓
Weights				✓

Notes: Panel A uses the difference-in-differences specification (equation 1) to estimate the effect of INPRES SD (defined as schools constructed per 1,000 children aged 5–14 years) on years of schooling, restricting the sample to male wage earners in the "young" to "old" cohorts, and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. Panel B uses the same specification and sample to estimate the effect of school construction on the log of monthly wages. Panel C uses the two-stage least squares specification (equation 3) where the interaction of school construction and the "young" cohort dummy instruments for years of schooling to estimate the effect of schooling on the log of hourly wages, again restricting the sample to male wage earners and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. Panel D uses the two-stage least squares specification (equation 4) where the interaction of birth-year dummies and school construction instrument for years of schooling to estimate the effect of schooling on the log of hourly wages, again restricting the sample to male wage earners and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. In Panels C and D, the 95% confidence sets were calculated using the wild bootstrap to invert the Anderson–Rubin test; and the F-statistic is the effective-first stage F from Montiel Olea and Pflueger (2013). Column (1) replicates the original from Duflo (2001) using the four specifications described above. Column (2) applies clustered standard errors at the level of the district of birth to those four specifications. Column (3) applies clustering and corrects the transcription errors identified by Roodman (2023). Column (4) applies clustering, corrects transcriptions errors, and incorporates survey weights. Panels A, B, and C all have the same number of observations.

3 "Macro" Comment: the "wage dilation" problem

The main concern of Roodman (2023) is a specific threat to the identification strategy for the reduced form impact of school building on wages. Suppose that the wage difference between educated and uneducated workers increases with age — for example, perhaps more educated workers select into occupations with potential promotions, which accrue over time; or, as described by Mincer (1974), the decreasing returns to experience kick in earlier for less educated workers because they are younger when enter the workforce. Then a difference-in-differences approach that compares younger to older cohorts would find that the wage gap between high- and low-education districts would be smaller in the younger cohorts than the older cohorts — more time has passed, allowing the more educated to accrue more benefits. Since more INPRES SD schools were assigned to districts with lower schooling levels to start with, even if the new schools had had zero impact on wages, we would still find a difference-in-differences effect: it would spuriously appear that the schools had caused the wages of the high- and low-education workers within the young cohort to converge, when in reality they simply have had less time to diverge. Roodman refers to this phenomenon as the "wage dilation problem."

Figure 1: Smoothed fits of hourly wage to age, by years of schooling, for wage-earning Indonesian men in 1995 (reproduced from Roodman (2023), Figure 3).



Roodman's Figure 3 (reproduced here as Figure 1) illustrates the problem. It shows smoothed fits of the age profiles of hourly wages for Indonesian men in 1995, separated into different education bins. Those with more education earn more, at any age, than those with less education — this could be causal or not. However, the wage gap between highly- and less highly-educated workers is less pronounced for younger workers than for older workers. Since, on average, workers in regions that got more schools were less educated, we would expect a positive difference in difference across cohort, even if INPRES did not affect schooling (or if this exogenously assigned schooling did not affect wages).

For example, Duflo (2001) compares the "young" cohort, which was 23–27 years old in 1995, to the "old" cohort," which is 33–38 years old. Looking again at Figure 1, suppose that those in the high–INPRES SD intensity (and, therefore, low-education) districts look like the teal line on average (7–9 years of schooling) while those in the low-intensity (high-education) districts look more like the blue line (10–12 years of schooling). Then a difference-

in differences specification would effectively compare the gap between the blue and the teal lines for 35-year-olds to the gap for 25-year-olds, and find that the gap has shrunk for the younger cohort — even absent any impact of INPRES SD.

Then why are the placebo tests in Duflo (2001) not subject to the same bias? Recall that to bolster her causal claim, Duflo included "control experiments" that compared two cohorts of which neither should have received any benefit from INPRES SD (see Duflo (2001), Tables 3 and 4). She fails to reject the null in any one of her many specifications of these control experiments, which would seem to refute Roodman's assertion that comparing older to younger cohorts across high- and low-intensity districts will necessarily lead to spurious difference-in-differences.

The null results in the placebo specifications do not, however, fully address Roodman's "wage dilation" hypothesis. The age gap in the experiment of interest is larger than in the placebo tests: the experiment of interest compares 23–27-year-olds to 33–38-year-olds, while the placebo experiment compares 33–38-year-olds to 39–45-year-olds. This means that in the placebo experiment, the high- and low-education regions have had a similar amount of time to diverge across the younger and the older cohorts. In summary, in the placebo tests, the dilation is diluted.

What, then, is the dilation solution? As Roodman points out, Figure 1 both diagnoses the problem and suggest a solution: at some point, when the workers are old enough, the age profiles become more parallel. For example, among those with 13–15 years of schooling, the income for 60-year-olds is even less than that of 55-year-olds; and, as a result, their income converges with that of less-educated 60-year-olds — i.e., the wage dilation has reversed. The obvious test of whether wage dilation caused a spurious positive coefficient in the reduced form specification is then to reproduce the original analysis with data from same cohorts as in Duflo (2001) but at an older age, by which point the wage dilation has dissipated; i.e., we need a later follow-up survey.

3.1 More Recent Follow-up Survey: SUSENAS

For this analysis, one needs a dataset more recent than 1995 that has education, wages, year of birth, and district of birth. The necessity of the first three variables is self-evident: it

would be impossible to study the effect of a program on education and wages differentially across years of birth without data on education, wages, and years of birth. District of birth — not just district of residence — also is essential because, as Hsiao (2023) shows, much of the effect of INPRES SD on wages is driven by migration: the program increased education and migration to areas with higher wages, which systematically received fewer INPRES schools. District of residence in adulthood therefore is a flawed proxy for district of residence in childhood and could downwardly bias a difference-in-differences estimate. These data requirements rule out two of the data sets proposed by Roodman (2023) for this analysis: the 2005 SUPAS does not have wages, and the 2010 SAKERNAS does not have district of birth. Only the SUSENAS meets the requirements, so here we only address analysis done with the SUSENAS.

First, to test whether the dilation problem is in fact reduced in the new data, Roodman regresses wages on schooling, age, and the interaction between schooling and age, restricting the sample to the relevant cohorts. What matters here is the coefficient on the interaction term: the dilation problem can be described as the more highly educated having a steeper slope in their wage–age curve. As he shows in Figure 5, the interaction term has a small but precisely estimated positive coefficient in this regression using the 1995 SUPAS data or the 2013–14 SUSENAS — but for the SUSENAS, the magnitude is cut in half, suggesting to Roodman that the dilation problem has dissipated but has not entirely disappeared. Therefore, it would appear that using the 2013–14 SUSENAS data is at least a partial solution to the wage dilation problem.

Roodman (2023) proceeds to redo the Duflo (2001) analysis using the 2013 and 2014 SUSENAS, presenting the results in Figure 5 and Table 3. He find null effects across the board: no significant effect of INPRES SD on years of schooling or on income (Figure 5) and no effect of education on income (Table 3). Given the lack of first stage, the 2SLS is uninterpretable, which is reflected in the large standard errors and infinitely-wide 95% confidence sets. At this point, it is inconclusive whether there are positive returns to education in this setting.

Coincidentally, Hsiao (2023) implement a similar analysis as Roodman (2023), using recent follow-up data to replicate Duflo (2001), with two major differences The first difference

is that Hsiao uses four waves of the SUSENAS (2011, 2012, 2013, and 2014) while Roodman uses only two. The second difference is that and Hsiao finds significant positive effects of the program on education and monthly wages, while Roodman finds null results. In fact, the magnitudes of Hsiao's point estimates very closely resemble those of Duflo (2001). What explains this discrepancy? Below, we examine the analyses of Roodman (2023) and Hsiao (2023) in detail in order to reconcile the differences.

We start by replicating the Roodman's results (see Roodman (2023), Table 3, Panel E, Columns (3) and (5)). We illustrate those results in Column (1) of Table 2, following the same layout that we do in Table 1. Panels A and B use the simple difference-in-differences specification (equation 1) to estimate the effect of primary school construction (defined at the district level as INPRES schools constructed per 1,000 children) on years of schooling and on the logarithm of monthly wages by comparing the "young" to the "old" cohort. We see here that although Roodman finds a borderline-significant first stage estimate of 0.114 years of schooling (only half the size of the 1995 first stage), the reduced form is negative and close to zero. This is reflected in Panel C, which follows equation 3 in using the interaction of school construction with the "young" indicator variable as an instrument for years of schooling. Panel D uses the many-instruments specification from equation 4, where an indicator for each birth year is interacted with school construction. This yields a positive but small and statistically indistinguishable from zero result.

3.1.1 Typos: Education

This first stage in Roodman specification, while low, is still too large. The 2013–14 SUSENAS asked about years of education in an indirect way. There are three separate questions: highest level of school ever *attended* (for which the response options are primary school, junior high school, etc.), number of years having attended that school, and highest school level *completed* (for which the response options are the same as the first). In order to stay faithful to the Duflo (2001) measure of education, which is "years of schooling," one must convert the highest school attended response to the minimum number of years required to get to that level (i.e., if someone's highest level attempted was high school, they must have completed junior high school, so they have at least 9 years of schooling) and add to that the number of years

attended. From his code, which he has helpfully posted to his GitHub, we believe that this is exactly how Roodman attempted to calculate years of schooling; however, it appears that he made a typo (see Appendix A). We correct this typo, which makes the first stage even weaker. [2]

3.1.2 Typos: Districts and Inflation

The Indonesian government often redraws its administrative units, splitting up districts, assigning them to new regions, etc. This means that the codes used to identify each district have changed substantially between when INPRES began and these follow-up surveys. To harmonize INPRES data and these modern surveys, researchers must determine which modern districts correspond to which past districts. Given the less-than-perfect documentation available for this tedious task, it is easy to make mistakes. We have thoroughly examined Roodman's harmonization and have done our best to correct what we believe are a few of oversights that result in a some observations to be accidentally dropped, some to duplicated, and some to be matched to the wrong district.[3]

In addition, because he uses SUSENAS waves from two different years, Roodman (2023) wisely adjusts the wages for inflation. Unfortunately, a typo in his code results in the 2013 adjustment not getting applied to the wages variable used in his regressions. In Column (3) of Table 2 we correct this typo and also apply what we believe to be the correct district reassignments. Comparing that to Column (2), we find that these small typos make but an imperceptible difference to the results.

2. A hawk-eyed reader may notice that the point estimates of the reduced form change between Columns (1) and (2) despite the fact that all that we have changed is our years of schooling variable: Roodman also codes those missing schooling as having zero years of education, when we code these as missing, they are dropped from the wage regression.

3. Among the districts that were dropped but that we believe should not have been are Tambrauw and Mamberamo Raya, which together include about 1,000 observations. We also believe that those born in Supiori should be classified as having been born in Biak Numfor rather than Sorong. Finally, Roodman identifies some districts used in the 2013–14 SUSENAS that he believes cannot be matched to only one INPRES district; e.g., those born in recently-created Sarmi was carved out of two existing districts: Maluku Utara and Yapen Waropen. To account for the fact either of those two districts may be the correct birthplace for somebody born in Sarmi, he creates two imputation observations, one for each district, and calculates imputation weights. But Roodman's code fails to make use of these weights, as a result, each observation from Sarmi assigned to Maluku Utara has an exact clone who was assigned to Yapen Waropen.

3.1.3 Self-employed

In these waves of the SUSENAS, there is an attempt made to measure the income of those who report being self-employed.[4] We have a few reasons to believe that these respondents should be excluded from this analysis. One reason is that the way in which their income is reported in the SUSENAS inexplicably differs from year to year — e.g., in 2012, all 73,000 respondents who report being self-employed also report zero income, despite the fact that, to the best of our knowledge, the questionnaire in 2012 is identical to that of 2011, 2013, and 2014. This should give us pause when considering whether to trust these observations.

Another reason, Duflo (2001) only examines wage earners in its main analyses; the 1995 SUPAS did not report income for the self-employed. Therefore, to be faithful to the original specifications, in Column (4) we drop all respondents who report being self-employed, which is a substantial change to the dataset. This moves the point estimate of the effect of school construction on wages from weakly negative to substantially (but noisily) positive. The search for a sufficiently strong first stage to be able to run an IV regression continues, however, as the first stage is unaffected by the removal of the self-employed respondents.

3.1.4 Additional SUSENAS Waves

As Roodman (2023) points out, Hsiao (2023) makes use of more than just the 2013 and 2014 SUSENAS: the 2011 and 2012 waves also contain all the necessary variables to run this analysis. Conversely, he does not use all of the 2013 and 2014 years, since two quarters were missing at the Harvard library. We were able to track down CDs containing those survey years at Harvard University's Lamont Library, purchased the missing quarter, and we run Roodman's cleaning procedure (with our aforementioned corrections applied) on the data, and incorporate it into the analysis. In Column (5), we report the results if we run the same regressions from Column (4) on just the 2011 and 2012 data. Here we find a significant effect of school construction on educational attainment (0.105 extra years of schooling for each extra INPRES school constructed per 1000 children) and on monthly income (a 2.5% increase), both of which are significant. This makes for an average return to education of

4. We define "self-employed" as anyone who reports being self-employed — whether unassisted, assisted by unpaid or temporary workers, or by paid or permanent workers — in their "main job."

23.5% as illustrated by the IV specification in Panel C, and of 15.7% in Panel D — although the effective F-statistics are still low, so we should focus on weak-instrument robust inference. The single-instrument estimate is positive, and larger than that in the original paper, with a confidence set entirely above zero. The Anderson–Rubin confidence intervals for the many instrument specification remain unbounded, though this could be due to the issues with AR confidence intervals and many instruments mentioned above. Even if one completely distrusts the instrumental variables specifications the reduced form results are significant and, if anything, larger than those in Duflo (2001).

The strong, positive results from 2011–12 (Column (5)) are somewhat in contrast to the more muted results from 2013–14 (Column (4)). The standard errors are unchanged, but the point estimates are quite a bit larger. The year by year analysis in the appendix shows that the first stage is largest in 2011. We have yet to find a satisfactory hypothesis that explains the differences across survey years, nor one that gives us reason to trust some years more than others, so we contend that the specification in which we should have the most confidence is that which combines all four SUSENAS waves, 2011–14. We report these results in Column (6); as expected, they approximately split the difference between the 2011–12 and the 2013–14 results. The first stage and reduced form are positive and moderately significant, implying that each extra INPRES school constructed per 1,000 children resulted in an increase in schooling by 0.085 years and in wages by 1.6%. Those combine to produce an estimated 19.4% return to one year of schooling, which is significant at the 5% level — although the AR confidence set (barely) includes zero. The many-instrument IV point estimate is also positive but smaller in magnitude and, again, suffers from weak instruments.

3.1.5 Weighting

As discussed above, we contend that including survey weights in these specifications is not appropriate. Nevertheless, in Column (7), we examine the effect of adding survey weights to our preferred specifications. In the weighted regression, there is no first stage. Perhaps coincidentally, the point estimate of the single-instrument IV regression in Panel C is unchanged compared to the unweighted regression, but thanks to the weakened first stage in the weighted specification, this result is entirely uninformative.

3.1.6 Other Follow-up Analyses

Note that the results for our preferred specification for the follow-up analysis, Column (6) of Table 2, while not dissimilar to those of Table 1 of Hsiao (2023), are far from identical. To examine the differences in detail, see Appendix Table A3, where we undertake the same exercise as we do for Roodman in Table 2. In brief, where Hsiao differs from us is that he uses a cruder (though perhaps less noisy) measure of years of schooling, restricts his sample to heads of household (for the sake of his migration-focused analysis), uses the INPRES data from Duflo (2001) that includes typos, matches a small number of modern districts to what we believe are the wrong historical districts, and is missing half of the 2014 SUSENAS (because it is unavailable at the Harvard Library; we purchased the rest from SILASTIK (Statistic Service Information System of Indonesia). These changes do not substantially affect the results, nor the conclusions that Hsiao (2023) draws from them.

Roodman (2023) points out that instead of years of schooling Duflo (2004) uses primary school completion as the first stage. An issue with this as an IV is that primary schooling isn't excludable if INPRES increases education among those who don't finish primary school (or increases secondary school completion among the always-takers), and if there are labor market returns to those additional years of schooling. We show primary completion as an alternative first stage in Appendix Table A2, which follows the same layout as Table 2. The first stages are consistently positive and highly significant, but the IV estimates, though also positive, are very imprecise.

3.1.7 Overall Conclusion from SUSENAS followup

We come away from this exercise somewhat more confident in the Duflo (2001) results. Correcting the minor issues in Roodman's analysis and doubling the sample size appears to have at least partially lifted the fog that had been brought on by the passage of time between 1995 and 2011–14. The first stage, as Roodman discusses, is likely subject to more attenuation bias over time (perhaps due to age heaping and confusion on district of birth), so it is no surprise that our estimate of the effect of INPRES SD on years of schooling is substantially smaller in 2011–14 than in 1995; in fact, its magnitude is about half as large.

Coupled with a reduced form estimate that is approximately the same magnitude in 2011–14 as in 1995, this means for an IV estimate of the returns to education being, if anything, larger in the recent follow-up, though the confidence interval includes the original study.

Although we should not take these point estimates literally, the results suggest, if anything, great persistence of the long-run effects of INPRES SD, leaving us less concerned about the wage dilation problem.

3.2 Other Analyses

Roodman also performs a variety of other extensions; his efforts to overlay splines to tell if there is a trend break in the education coefficients and to examine all quantiles of the treatment effects in a changes-in-changes framework are interesting but push the data farther than it is powered to go. The original paper did not find that the 2x2 DiD results were significant, and, because changes-in-changes can't accommodate a continuous treatment variable, those analyses of quantile treatment effects are unfortunately underpowered.[5] Similarly, treating the coefficient estimates as a time series in which to detect breaks is difficult due to their imprecision: they are already estimated parameters, and each individual year coefficient is estimated with some amount of noise.

5. It is also a little difficult to state the rank invariance assumption for the changes-in-changes analysis given the underlying continuous treatment variable: some of the treated individuals were treated by more INPRES schools than others; for their ranking to be stable in the absence of treatment would require a sharp null that INPRES schools have no effect on anyone's wages. The rank invariance assumption could also be violated by systematic crossing of life-cycle wage profiles (see, e.g. Mello, Nybom, and Stuhler (2022).

Table 2: Re-analyzing Roodman (2023)

Outcomes	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First Stage (instrument: school construction interacted with "young" cohort dummy)							
Years of	0.114	0.065	0.062	0.061	0.105*	0.085*	0.033
School	(0.065)	(0.040)	(0.041)	(0.049)	(0.049)	(0.041)	(0.052)
Panel B: Reduced Form (instrument: school construction interacted with "young" cohort dummy)							
ln(Wages)	−0.001	−0.003	−0.003	0.008	0.025*	0.016	0.006
	(0.007)	(0.007)	(0.007)	(0.010)	(0.011)	(0.009)	(0.011)
Panel C: 2SLS (instrument: school construction interacted with "young" cohort dummy)							
ln(Wages)	−0.009	−0.044	−0.057	0.135	0.235**	0.194*	0.186
	(0.065)	(0.125)	(0.138)	(0.157)	(0.090)	(0.085)	(0.252)
95% CI	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[0.051, 1.570]$	$[-0.051, 2.124]$	$[-\infty, \infty]$
F-Stat	3.131	2.600	2.291	1.555	4.476	4.169	0.413
Obs.	125047	122871	122652	61332	56840	118162	118162
Panel D: 2SLS (instruments: school construction interacted with birth-year dummies)							
ln(Wages)	0.024	0.025	0.007	0.043	0.157**	0.082	0.045
	(0.029)	(0.044)	(0.048)	(0.066)	(0.056)	(0.051)	(0.058)
95% CI	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$
F-Stat	2.054	1.864	1.728	0.953	0.181	1.695	0.194
Obs.	221041	216574	216207	102909	95123	198032	198032
Correct Schooling		✓	✓	✓	✓	✓	✓
Correct Districts, Inflation			✓	✓	✓	✓	✓
Drop Self-Employed				✓	✓	✓	✓
2011–12					✓	✓	✓
2013–14	✓	✓	✓	✓		✓	✓
Weights							✓

Notes: Panel A uses the difference-in-differences specification (equation 1) to estimate the effect of INPRES SD (defined as schools constructed per 1,000 children aged 5–14 years) on years of schooling, restricting the sample to male wage earners in the "young" to "old" cohorts, and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. Panel B uses the same specification and sample to estimate the effect of school construction on the log of monthly wages. Panel C uses the two-stage least squares specification (equation 3) where the interaction of school construction and the "young" cohort dummy instruments for years of schooling to estimate the effect of schooling on the log of hourly wages, again restricting the sample to male wage earners and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. Panel D uses the two-stage least squares specification (equation 4) where the interaction of birth-year dummies and school construction instrument for years of schooling to estimate the effect of schooling on the log of monthly wages, again restricting the sample to male wage earners and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. In Panels C and D, the 95% confidence sets were calculated using the wild bootstrap to invert the Anderson–Rubin test; and the F-statistic is the effective-first stage F from Montiel Olea and Pflueger (2013). Column (1) replicates the Panel E of Table 3 from Roodman (2023), running the four specifications described above on data from the 2013 and 2014 waves of the SUSENAS. Column (2) corrects the typo in Roodman's code that creates the "Years of Schooling" variable. Column (3) corrects what we believe are mistakes in matching SUSENAS-era district identifiers to their INPRES-era equivalents, and a typo in the code that adjusts incomes for inflation. Column (4) drops those who report being self-employed (for whom income is inconsistently calculated). Column (5) restricts the data to the 2011 and 2012 waves of the SUSENAS. Column (6) combines the 2011–14 SUSENAS waves. Column (7) applies survey weights.

4 Conclusion

Roodman (2023) proposes a close examination of the Duflo (2001) study of the impact of the INPRES school construction program on education and labor market outcomes. He corrects typos that have affected not only Duflo's original results but also those of the growing list of papers than have taken advantage of her INPRES data. These corrections, don't turn out to be important. Roodman also points out that the first stage is not particularly strong, and modern IV techniques lead to higher standard errors than originally reported.

More substantively, Roodman also suggests a reason for a a possible violation of parallel trends assumption on which Duflo (2001) relies: in the cross-section, the education wage gap grows with age, which would upwardly bias Duflo's difference-in-differences regressions. Roodman proposes several technical solutions to this "dilation problem," none of which prove fully satisfactory. His more promising solution is to follow up with the original cohorts later in life, by which point the "dilation problem" should have dissipated; however, mainly due to a lack of suitable data (with a first stage) , Roodman finds little conclusive evidence, one way or another. When we add to his data set, doubling the sample size, we find results that echo those of Duflo (2001).

Overall, we come away from this exercise with more evidence that the INPRES school construction program did in fact appear to increase schooling and earnings, with impacts on earnings that are strikingly persistent, 40 years after the schools were constructed and the first students attended. Obviously, this is but one study, which never had ahuge degree of precision, and it would not be reasonable to rely heavily on this particular data point to extrapolate rates of returns to education in developing countries. Fortunately, the literature on the impacts of education is continuing to grow.

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A Appendix: Roodman's typo in the construction of the education variable in the SUSENAS

The SUSENAS does not directly give years of education. Instead, it asks people what level they attended, and how many years they attended at the next level (if they did not complete, where 8 means they completed).

Roodman's code reads:

```
recode B5R15 (1 2 3 = 0) (4 5 6 = 6) (7 8 9 10 = 9) (11 12 13 14 = 12),
gen(yeduc)  // years of schooling *before* each schooling level
    mat completionyears = 6, 6, 6, 3, 3, 3, 3, 3, 3, 3, 3, 2, 3, 5
    // max years in schooling levels; used when p518=8, meaning "completed"
    replace yeduc = yeduc + cond(B5R15<8, B5R15,
            completionyears[1, B5R15]) if B5R15>0.
```

We believe that Roodman's intent for the last was:

```
    replace yeduc = yeduc + cond(B5R16<8, B5R16,
            completionyears[1, B5R15]) if B5R15>0.
```

where B5R15 is the highest level of education attempted and B5R16 is the number of years at that level (so if someone attempted primary but dropped out in the first year, it would be BR15=1, BR16=1, and years of education should be zero).

For ease of reference, here is our code, which mostly attempts to do what Roodman attempts to do, minus the typo. The only conceptual difference is that we don't give a year of credit for attempted years (so if someone attempted 1 year of a higher level, we give a credit of zero).

```
rename B5R15 school_attempted
rename B5R16 attempted_years
generate years_schooling_base = school_attempted
recode years_schooling_base (1 2 3 = 0) (4 5 6 = 6) (7 8 9 10 = 9) (11 12 13
generate years_schooling_add = attempted_years - 1 if attempted_years >= 1
replace years_schooling_add = 0 if attempted_years == 0
```

```
replace years_schooling_add = 6 if inrange(school_attempted, 1, 3) & attempte
replace years_schooling_add = 3 if (inrange(school_attempted, 4, 10) | school_
replace years_schooling_add = 1 if school_attempted == 11 & attempted_years =
replace years_schooling_add = 0 if school_attempted == 14 & (attempted_years
replace years_schooling_add = 2 if school_attempted == 14 & attempted_years =
generate years_schooling_correct = years_schooling_base + years_schooling_add
```

B Appendix

Appendix Table A1: Reduced Form by Year

Outcomes	2011	2012	2013	2014
Panel A: First Stage, Roodman approach				
Years of	0.108	0.135	0.112	0.121
School	(0.075)	(0.106)	(0.072)	(0.081)
Panel B: Reduced Form, Roodman approach				
ln(Wages)	0.011	0.030*	−0.004	0.002
	(0.009)	(0.013)	(0.010)	(0.010)
Obs	69933	25054	56938	68109
Panel C: First Stage, with our "micro" corrections				
Years of	0.113	0.072	0.047	0.080
School	(0.065)	(0.065)	(0.058)	(0.068)
Panel D: Reduced Form, with our "micro" corrections				
ln(Wages)	0.018	0.028*	0.005	0.013
	(0.014)	(0.013)	(0.012)	(0.013)
Obs	32236	24604	29087	32235

Notes: Our "micro" corrections are to correct the typo in Roodman's code that creates the "Years of Schooling" variable, correct what we believe are mistakes in matching SUSENAS-era district identifiers to their INPRES-era equivalents, to adjust incomes for inflation, and to drop those who report being self-employed.

Appendix Table A2: The Effect of Primary School Completion on Earnings

Outcomes	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First Stage (instrument: school construction interacted with "young" cohort dummy)							
Completed	0.013*	0.016***	0.016***	0.013**	0.016**	0.014***	0.021***
Primary	(0.005)	(0.004)	(0.004)	(0.005)	(0.005)	(0.004)	(0.005)
Panel B: Reduced Form (instrument: school construction interacted with "young" cohort dummy)							
ln(Wages)	−0.001	−0.001	−0.002	0.011	0.028**	0.019*	0.012
	(0.007)	(0.007)	(0.007)	(0.011)	(0.011)	(0.009)	(0.011)
Panel C: 2SLS (instrument: school construction interacted with "young" cohort dummy)							
ln(Wages)	−0.086	−0.068	−0.110	0.835	1.787*	1.337*	0.549
	(0.591)	(0.466)	(0.473)	(0.867)	(0.776)	(0.653)	(0.479)
95% CI	$[-2.86, 1.27]$	$[-1.50, 0.84]$	$[-1.60, 0.80]$	$[-1.26, 4.55]$	$[0.43, 5.27]$	$[0.12, 3.86]$	$[-0.66, 1.59]$
F-Stat	6.343	13.354	13.048	7.245	11.039	12.390	17.033
Obs.	125047	125047	124835	61944	57831	119775	119775
Panel D: 2SLS (instruments: school construction interacted with birth-year dummies)							
ln(Wages)	0.114	−0.028	−0.088	0.264	1.792*	0.785	0.392
	(0.342)	(0.337)	(0.347)	(0.446)	(0.792)	(0.465)	(0.368)
95% CI	$[-\infty, \infty]$	$[-117455, \infty]$	$[-6.88, 15.3]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-1.84, 10.08]$	$[-10.58, 3.07]$
F-Stat	1.838	0.474	0.458	0.627	0.359	0.471	0.933
Obs.	221041	221041	220689	104183	97271	201454	201454
Correct Schooling		✓	✓	✓	✓	✓	✓
Correct Districts, Inflation			✓	✓	✓	✓	✓
Drop Self-Employed				✓	✓	✓	✓
2011–12					✓	✓	✓
2013–14	✓	✓	✓	✓		✓	✓
Weights							✓

Notes: Panel A uses the difference-in-differences specification (equation 1) to estimate the effect of INPRES SD (defined as schools constructed per 1,000 children aged 5–14 years) on likelihood of completing primary school, restricting the sample to male wage earners in the "young" to "old" cohorts, and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. Panel B uses the same specification and sample to estimate the effect of school construction on the log of monthly wages. Panel C uses the two-stage least squares specification (equation 3) where the interaction of school construction and the "young" cohort dummy instruments for primary school completion to estimate the effect of completing primary school on the log of hourly wages, again restricting the sample to male wage earners and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. Panel D uses the two-stage least squares specification (equation 4) where the interaction of birth-year dummies and school construction instrument for primary school completion to estimate the effect of competing primary school on the log of monthly wages, again restricting the sample to male wage earners and controlling for the interaction of birth-year dummies interacted with district population aged 5–14. In Panels C and D, the 95% confidence sets were calculated using the wild bootstrap to invert the Anderson–Rubin test; and the F-statistic is the effective-first stage F from Montiel Olea and Pflueger (2013). Column (1) replicates the Panel E of Appendix Table A-2 from Roodman (2023), running the four specifications described above on data from the 2013 and 2014 waves of the SUSENAS. Column (2) corrects the typo in Roodman's code that creates the "Years of Schooling" variable. Column (3) corrects what we believe are mistakes in matching SUSENAS-era district identifiers to their INPRES-era equivalents, and a typo in the code that adjusts incomes for inflation. Column (4) drops those who report being self-employed (for whom income is inconsistently calculated). Column (5) restricts the data to the 2011 and 2012 waves of the SUSENAS. Column (6) combines the 2011–14 SUSENAS waves. Column (7) applies survey weights.

Appendix Table A3: Micro Corrections to Hsiao (2023)

Outcomes	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First Stage (instrument: school construction interacted with "young" cohort dummy)							
Years of	0.121*	0.080	0.076	0.048	0.046	0.061	0.085*
School	(0.049)	(0.055)	(0.057)	(0.049)	(0.047)	(0.042)	(0.041)
Panel B: Reduced Form (instrument: school construction interacted with "young" cohort dummy)							
ln(Wages)	0.020*	0.017	0.017	0.017	0.014	0.013	0.016
	(0.009)	(0.010)	(0.010)	(0.010)	(0.010)	(0.009)	(0.009)
Panel C: 2SLS (instrument: school construction interacted with "young" cohort dummy)							
ln(Wages)	0.161*	0.210	0.218	0.347	0.295	0.218	0.194*
	(0.062)	(0.115)	(0.127)	(0.250)	(0.216)	(0.128)	(0.085)
95% CI	$[0.005, 0.545]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-0.051, 2.125]$
F-Stat	6.013	2.132	1.789	0.952	0.966	2.112	4.169
Obs.	89404	89238	89201	89201	102905	115195	118162
Panel D: 2SLS (instruments: school construction interacted with birth-year dummies)							
ln(Wages)	0.103*	0.107*	0.116*	0.146**	0.104	0.088	0.082
	(0.041)	(0.049)	(0.048)	(0.055)	(0.054)	(0.061)	(0.051)
95% CI	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$	$[-\infty, \infty]$
F-Stat	0.538	0.368	0.388	0.235	0.226	0.173	0.255
Obs.	149533	149208	149195	149195	172342	193206	198032
Correct Typos		✓	✓	✓	✓	✓	✓
Correct Districts, Inflation			✓	✓	✓	✓	✓
Redefine Schooling				✓	✓	✓	✓
Include 2014 Q2, Q4					✓	✓	✓
Include non-Heads of Household, Unemployed						✓	✓
Only Basic Controls							✓

Notes: Panel A uses the difference-in-differences specification (equation 1) to estimate the effect of INPRES SD (defined as schools constructed per 1,000 children aged 5–14 years) on years of schooling, restricting the sample to male wage earners in the "young" to "old" cohorts. Panel B uses the same specification and sample to estimate the effect of school construction on the log of monthly wages. Panel C uses the two-stage least squares specification (equation 3) where the interaction of school construction and the "young" cohort dummy instruments for years of schooling to estimate the effect of schooling on the log of hourly wages, again restricting the sample to male wage earners. Panel D uses the two-stage least squares specification (equation 4) where the interaction of birth-year dummies and school construction instrument for years of schooling to estimate the effect of schooling on the log of monthly wages, again restricting the sample to male wage earners. In Panels C and D, the 95% confidence sets were calculated using the wild bootstrap to invert the Anderson–Rubin test; and the F-statistic is the effective-first stage F from Montiel Olea and Pflueger (2013). Column (1) replicates Table 1 of Hsiao (2023), running the four specifications described above — including survey-year fixed effects, and controlling for interactions of the "young" cohort dummy with three different district-specific variables: population of children age 5–14, fraction of children age 5–14 enrolled in school, and the INPRES water and sanitation program spending per capita — on the 2011, 2012, 2013 waves of the SUSENAS, plus the first and third quarters of 2014, restricted to employed male heads of households. Column (2) corrects the transcription errors in the INPRES data of Duflo (2001) identified by Roodman (2023). Column (3) corrects what we believe are mistakes in matching SUSENAS-era district identifiers to their INPRES-era equivalents, and adjusts incomes for inflation. Column (4) modifies the calculation of years of schooling to adhere to what we believe is a more precise definition. Column (5) adds the second and fourth quarters of the 2014 SUSENAS. Column (6) removes the restrictions that limited the sample to those who report being employed heads of the household. Column (7) replaces the survey-year fixed effects and the interactions of the "young" cohort dummy with the vector of district-specific controls with interactions of birth-year dummies with the population of children aged 5–14 in the district in 1971.